

SkyVerse AI - Development Project Plan (v1)

Goal: Build a single React + Three.js application that first works on a phone in AR mode (0 cost) and later supports Ceiling Projection mode using a projector without changing the rendering engine.

Core Design Principle

One SkyScene. Every feature (AR, Ceiling, Weather, Educational, Minimal, Time Travel) reuses the same Three.js SkyScene. Modes only change camera behavior, overlays and UI.

Recommended Modules

Module	Purpose	Technology
SkyScene	Shared Three.js celestial scene	Used by every mode
Mode Manager	Switch AR/Ceiling/Weather/etc.	Controls visibility & camera
Astronomy Service	Star, Moon & planet positions	astronomy-engine
Weather Service	Clouds, rain, wind	OpenWeather or WeatherAPI
Satellite Service	ISS, Starlink	satellite.js + TLE data
Sensor Service	GPS, compass, orientation	Browser Geolocation + DeviceOrientation
Voice Service	'Hey Projector' commands	Web Speech API
Camera Service	Phone camera	MediaDevices API
UI Layer	Menus, labels, settings	React

Architecture Flow

GPS + Time + Device Orientation -> Astronomy Service -> SkyScene. Weather API -> Weather Layer. Satellite Service -> Satellite Layer. Voice Service -> Mode Manager. Mode Manager configures the shared SkyScene for AR or Ceiling output.

Suggested APIs

- **astronomy-engine:** Real star, Moon and planet positions
- **OpenWeather / WeatherAPI:** Weather, rain, cloud cover
- **satellite.js:** Satellite calculations
- **Browser Geolocation API:** Latitude/Longitude
- **DeviceOrientation API:** Compass & phone orientation
- **MediaDevices.getUserMedia:** Camera feed for AR
- **Web Speech API:** Voice commands

Development Phases

Phase	Name	Deliverable
1	Setup	React, Vite, folder structure, routing.
2	SkyScene	Three.js + React Three Fiber shared scene.
3	Astronomy	Integrate astronomy-engine.

4	AR Mode	Camera + GPS + orientation.
5	Weather	Integrate weather API and cloud overlay.
6	Satellites	ISS and Starlink.
7	Modes	Planetarium, AR, Ceiling, Weather, Educational, Minimal, Time Travel.
8	Voice	'Hey Projector' command parser and mode switching.
9	Testing	Phone testing, performance, battery.
10	Projection	Cast to projector; enable Ceiling mode.

Estimated Cost

Development (Phone MVP): ■0. Later Projection: Mini projector ■3,000–■5,000 (+ optional HDMI/casting adapter).

AI Prompting Tip

Use this document as the source of truth when asking AI/GHCP to generate code. Request one module at a time (e.g., 'Generate the SkyScene module using React Three Fiber and astronomy-engine'). Avoid asking for the entire app in one prompt.